STUDENT LABORATORY MANUAL

Module 5: Introduction to Computer Graphics and Visual Computing

20 Guided Examples • 20 Practice Exercises • 20 Outputs

Course: CS0045 – Computer Graphics and Visual Computing

Companion: Module 5 Student Guide

Toolchain: VS Code + MSYS2 UCRT64 GCC + GLEW + FreeGLUT

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Section:	**TN35**
Professor:	**Dr. Hadji J. Tejuco**

GitHub Repository: https://github.com/nicoleanneebron-art/Ebron-Nicole-Anne-V.

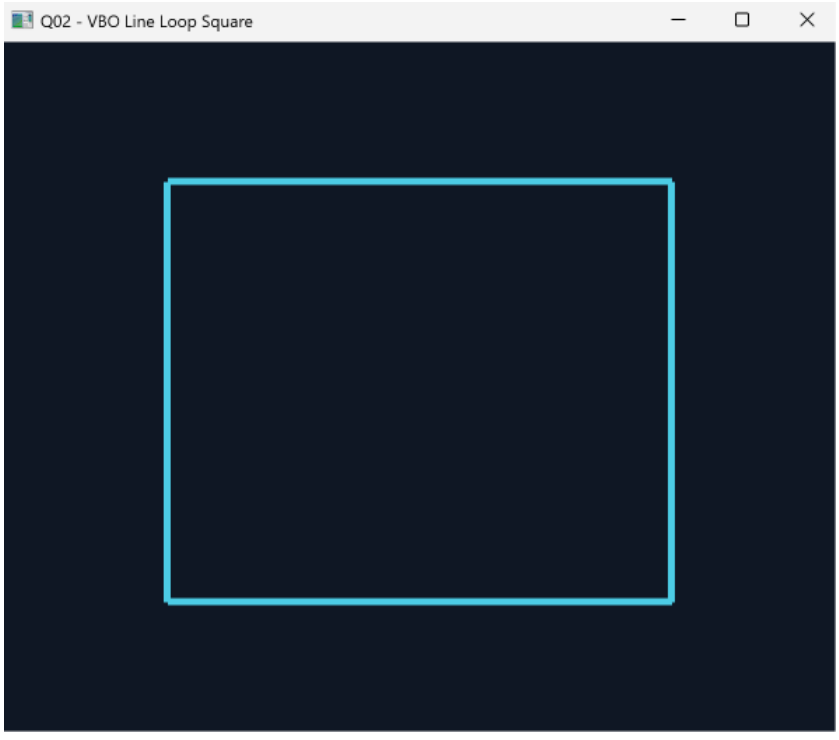
Part C — Outputs / Screen Shots (Ex01–Ex20)

Easy Exercises

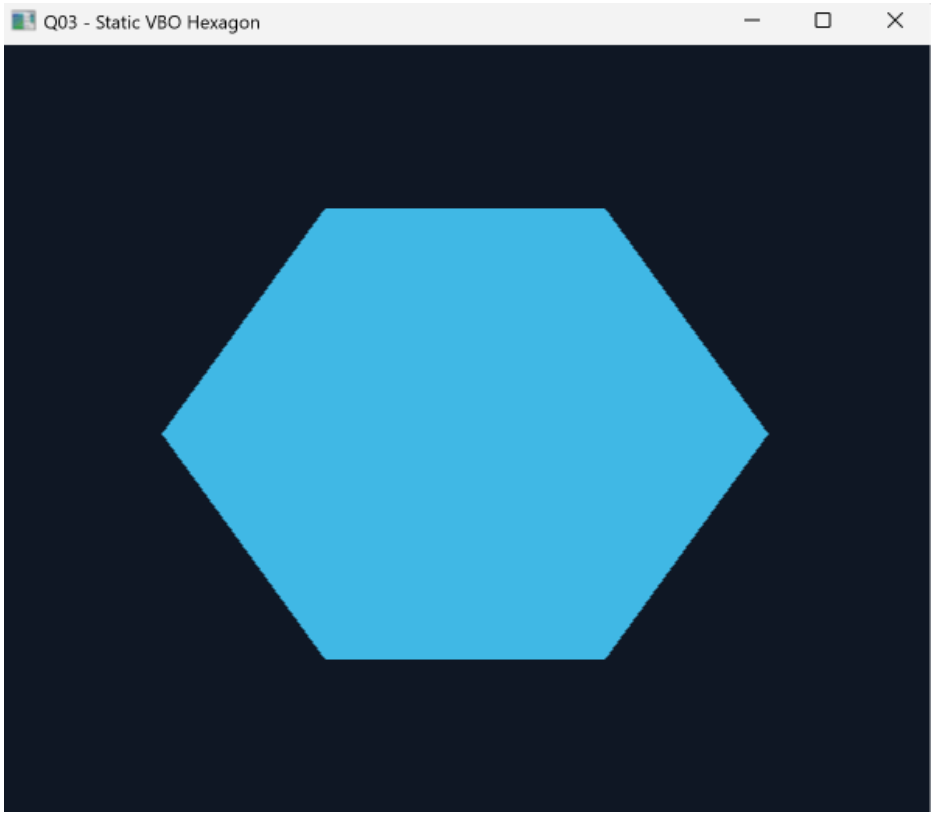
Exercise Q01 — VBO X-Pattern Points [Easy]



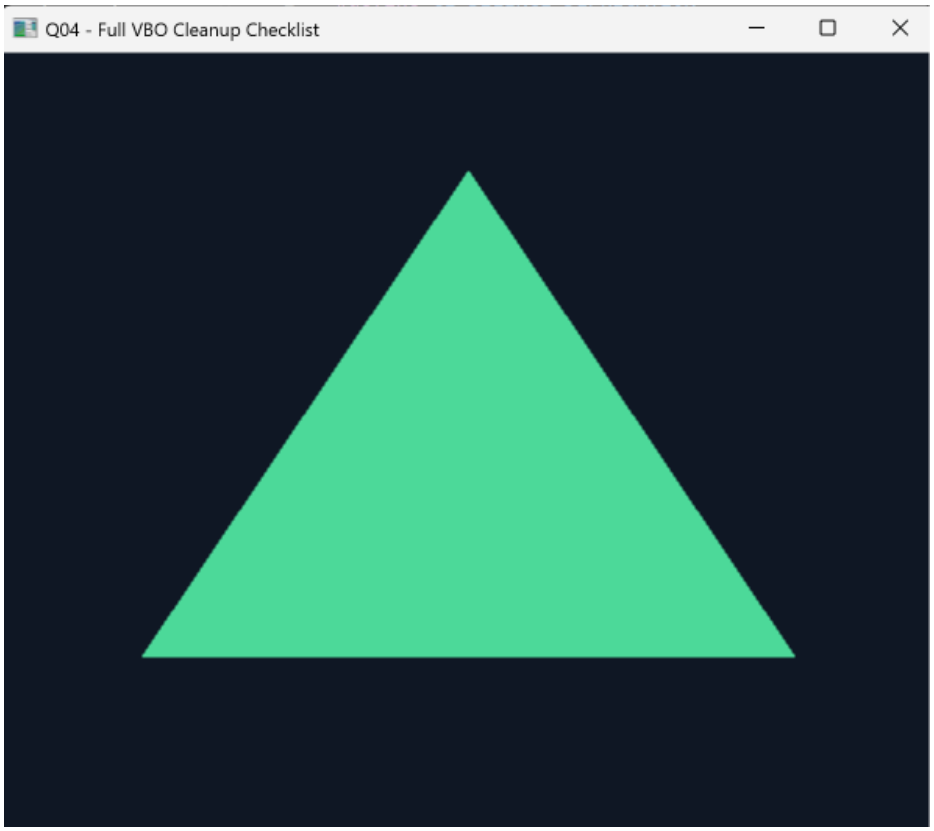
Exercise Q02 — VBO Line Loop Square [Easy]



Exercise Q03 — Static Hexagon with the Correct Usage Hint [Easy]



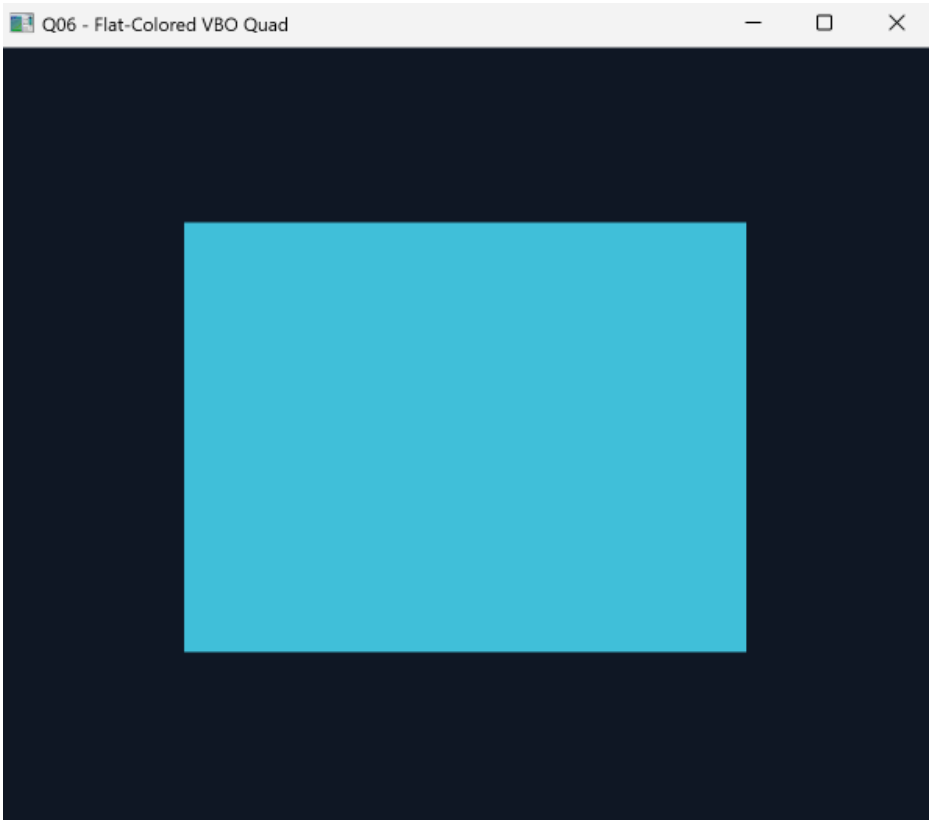
Exercise Q04 — Full Cleanup Checklist [Easy]



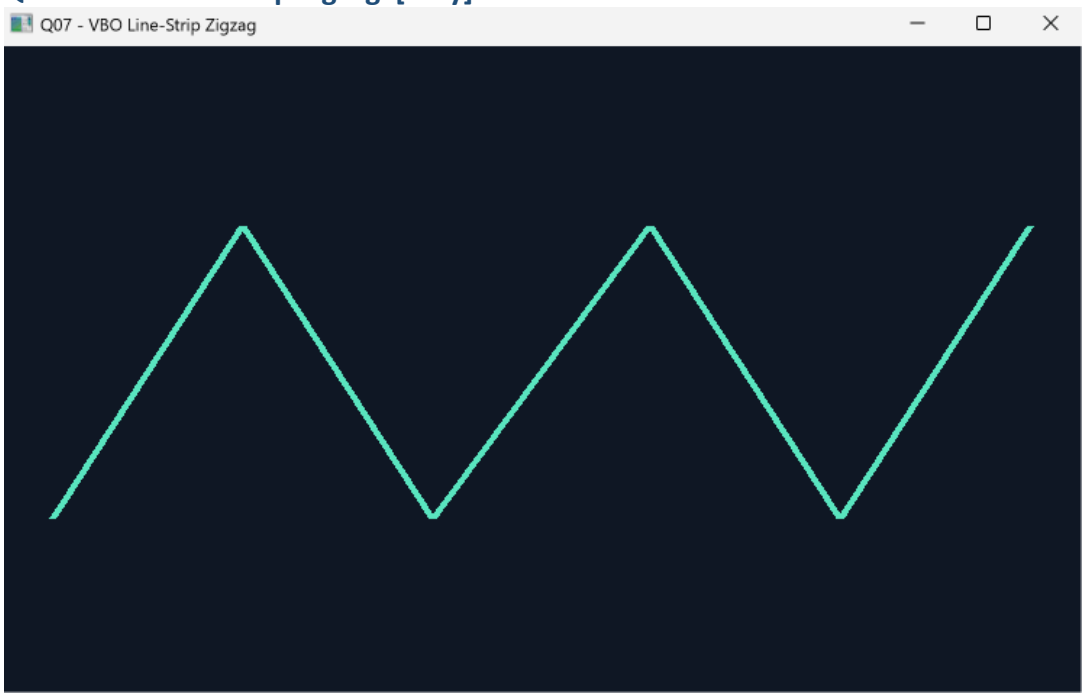
Exercise Q05 — Two Separate VBOs Side by Side [Easy]



Exercise Q06 — Flat-Colored VBO Quad [Easy]

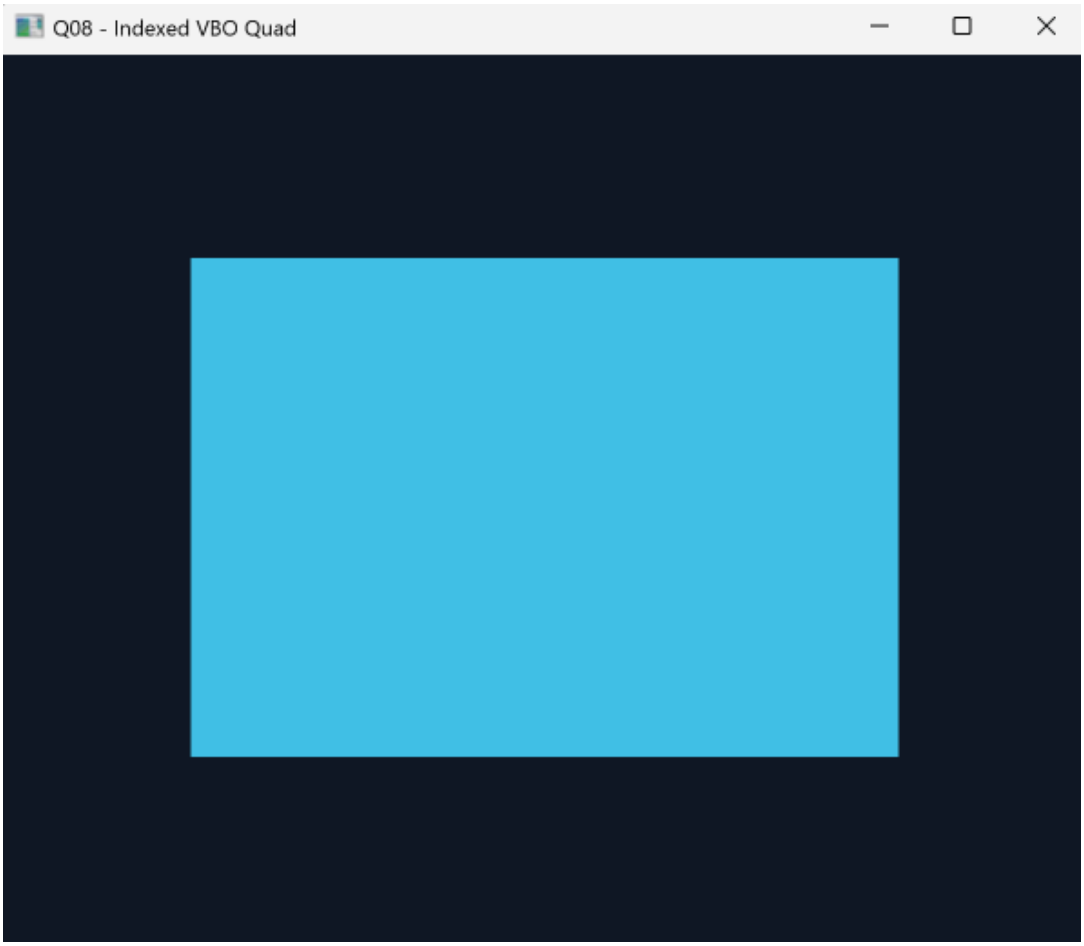


Exercise Q07 — VBO Line-Strip Zigzag [Easy]

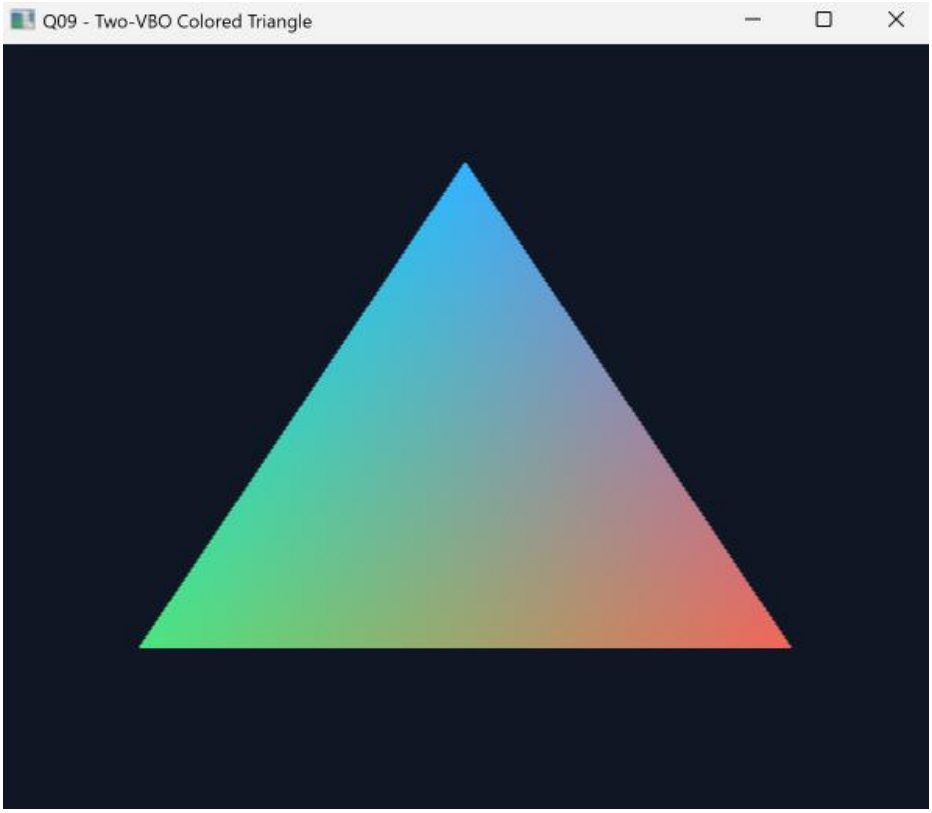


Medium Exercises

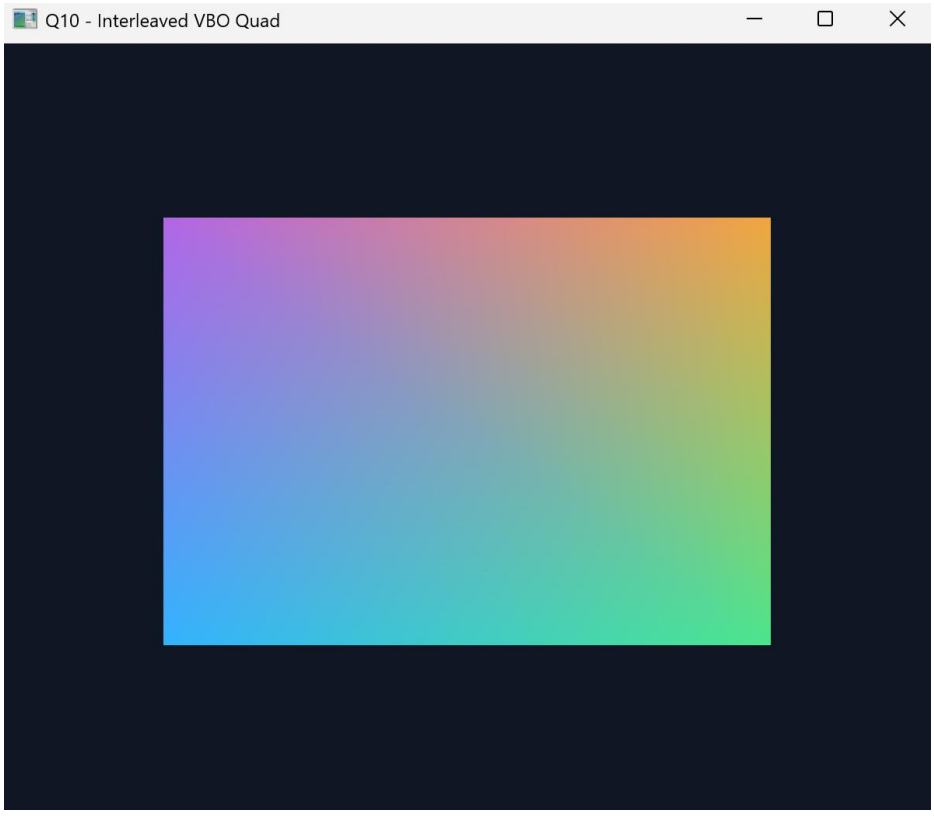
Exercise Q08 — Indexed Quad: Position VBO + Separate Index VBO [Medium]



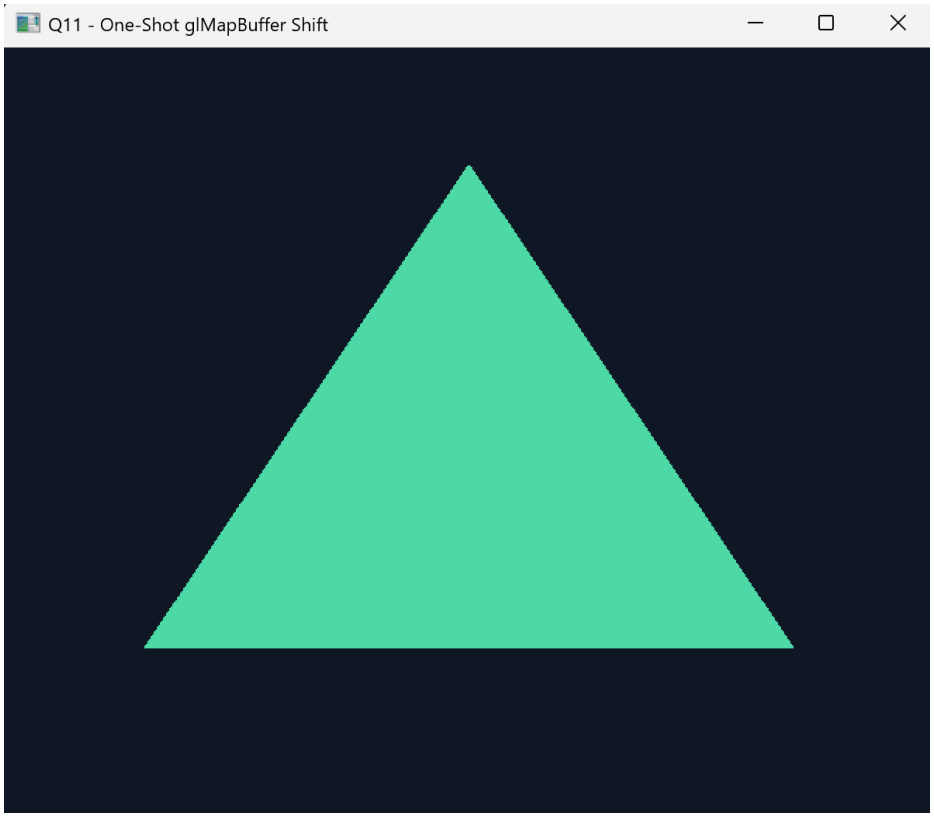
Exercise Q09 — Two-VBO Colored Triangle [Medium]



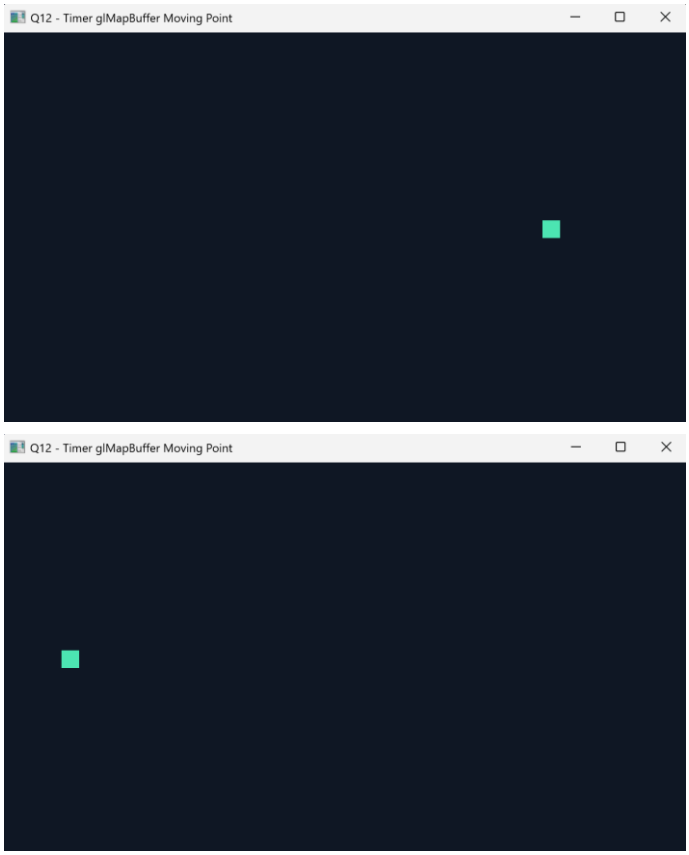
Exercise Q10 — Interleaved VBO Quad [Medium]



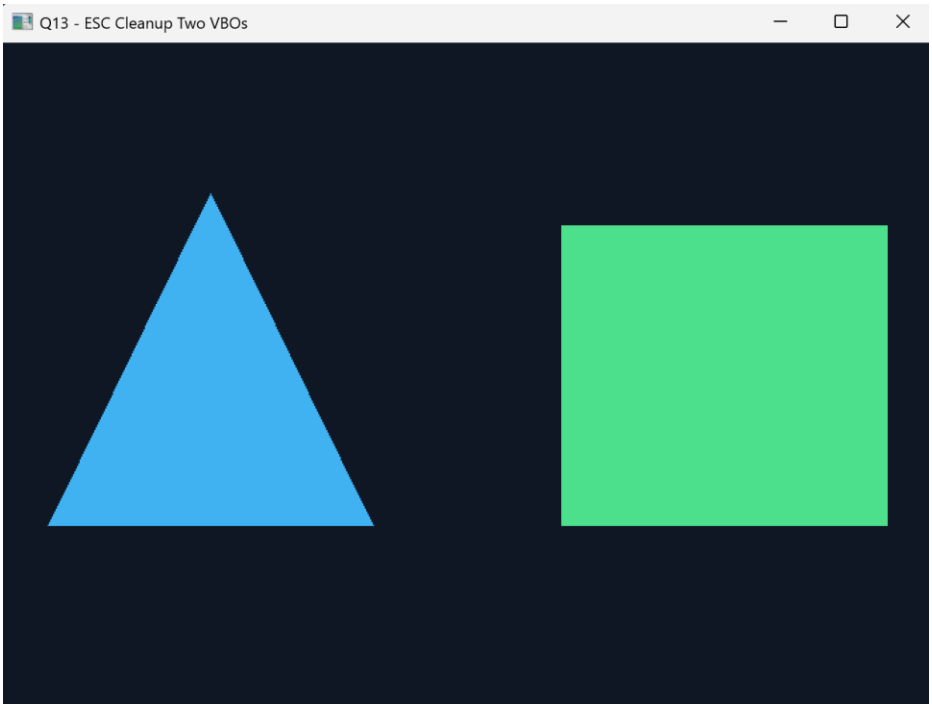
Exercise Q11 — One-Shot glMapBuffer Vertical Shift [Medium]



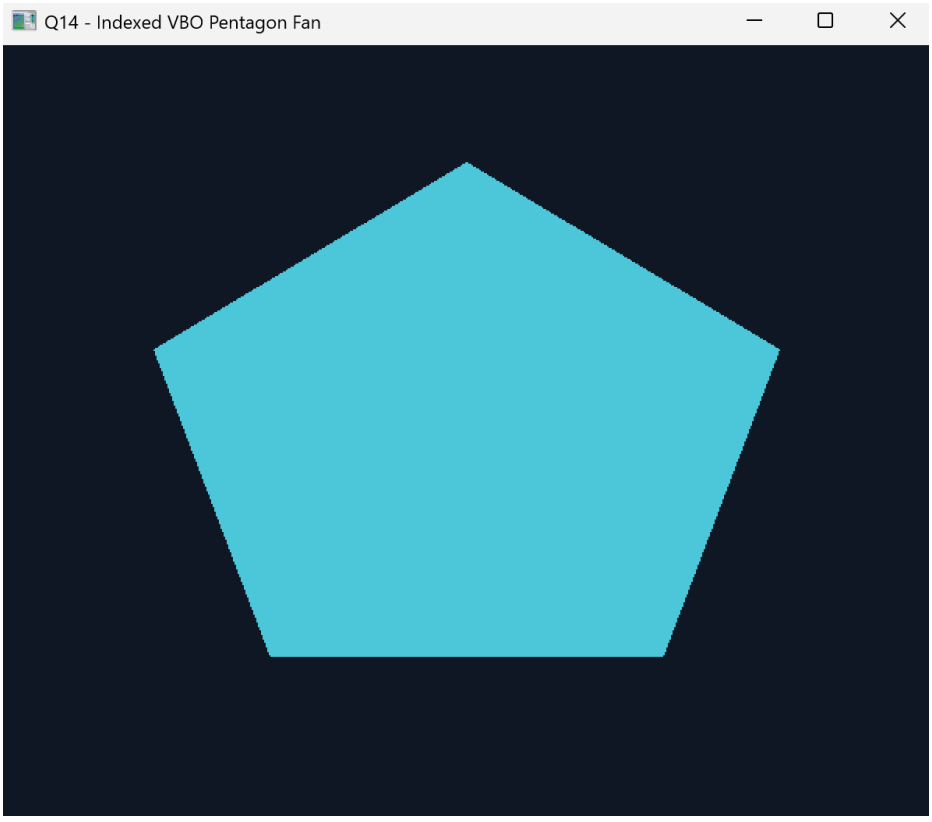
Exercise Q12 — Timer + glMapBuffer Moving Point [Medium]



Exercise Q13 — ESC Cleanup Releasing Two VBOs [Medium]

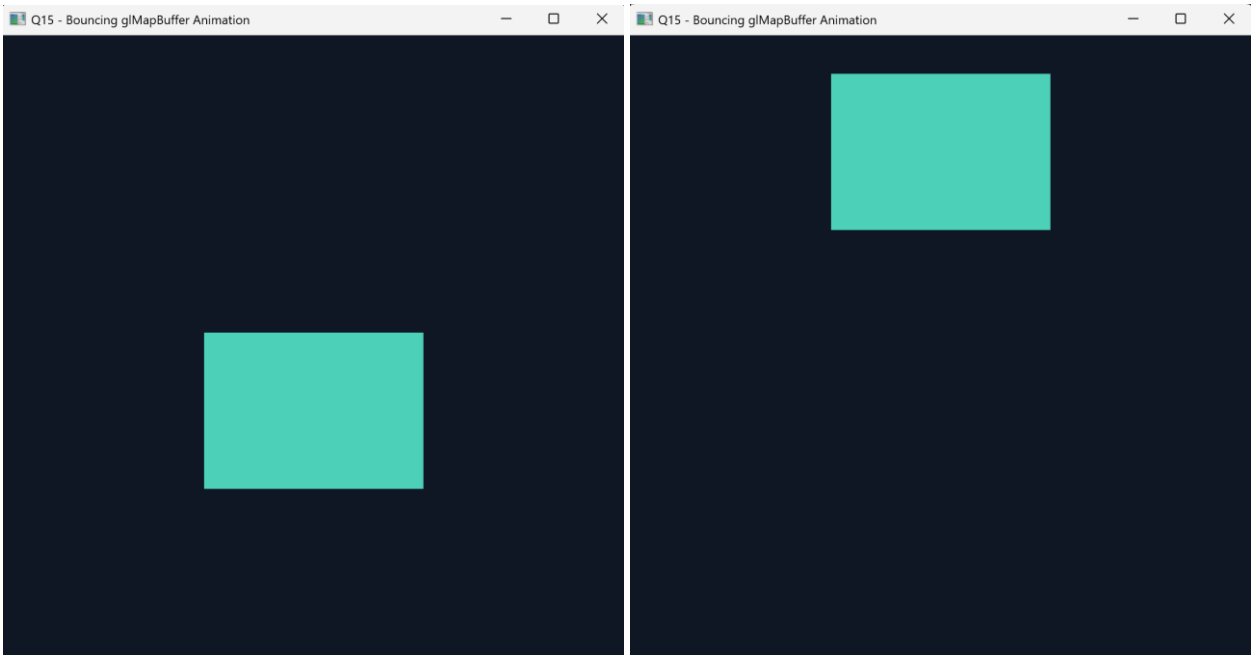


Exercise Q14 — Indexed GL_TRIANGLE_FAN Pentagon [Medium]

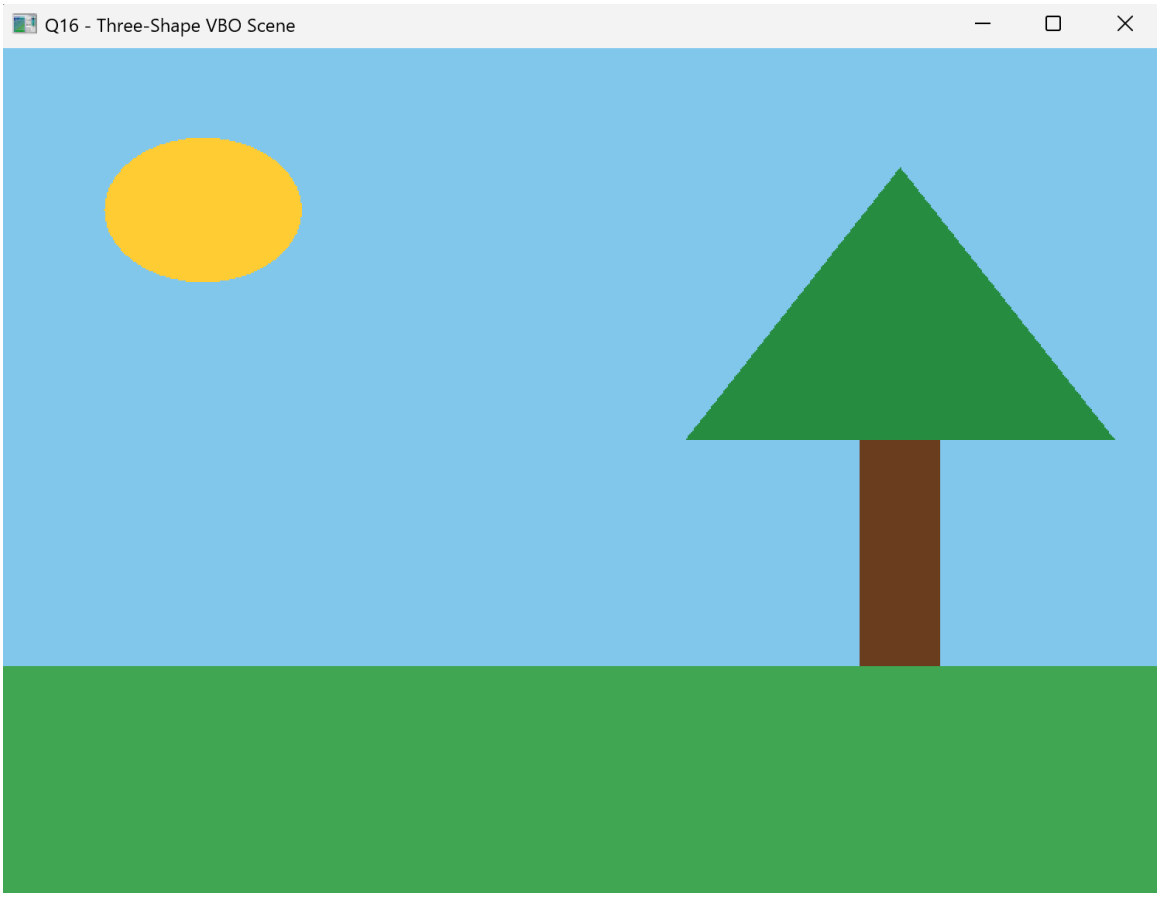


Hard Exercises

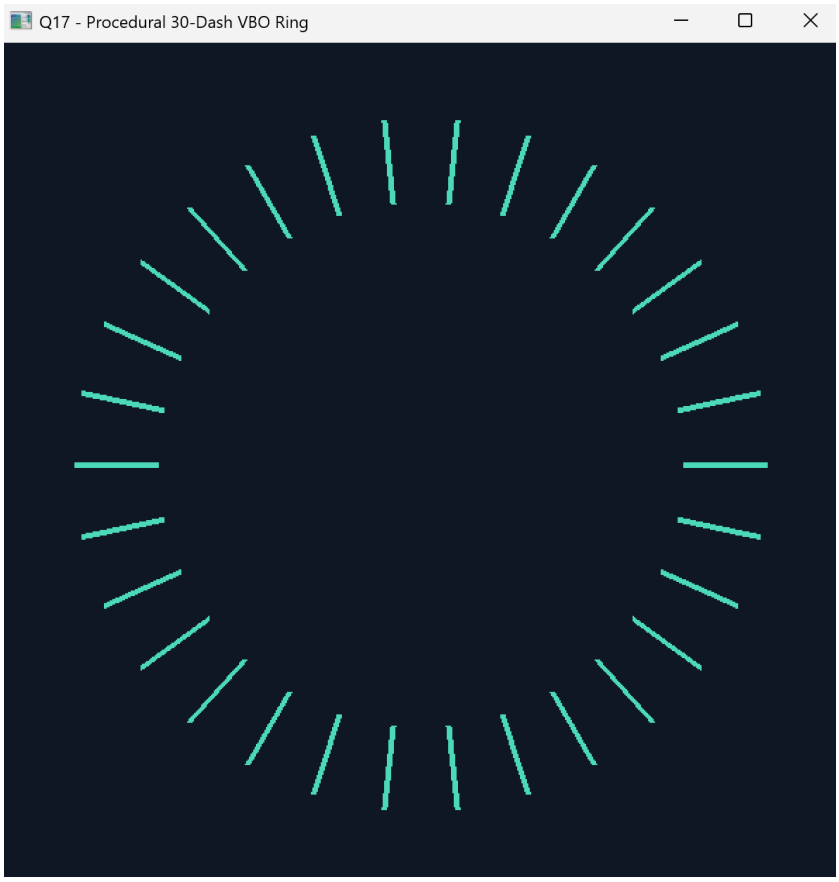
Exercise Q15 — Continuous Bouncing glMapBuffer Animation [Hard]



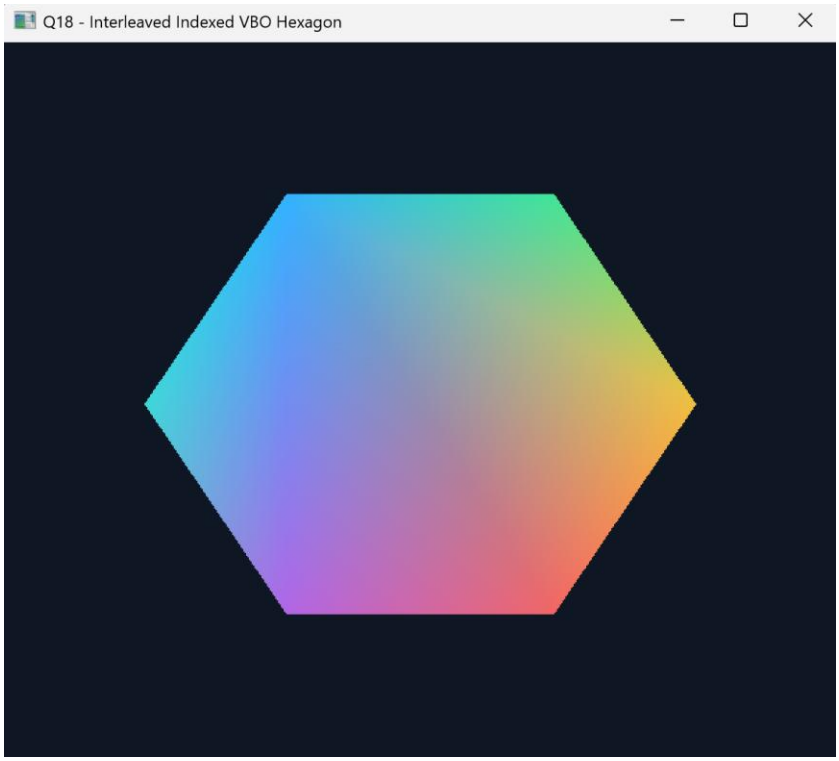
Exercise Q16 — Three-Shape VBO Scene: Sun, Indexed Ground, Tree [Hard]



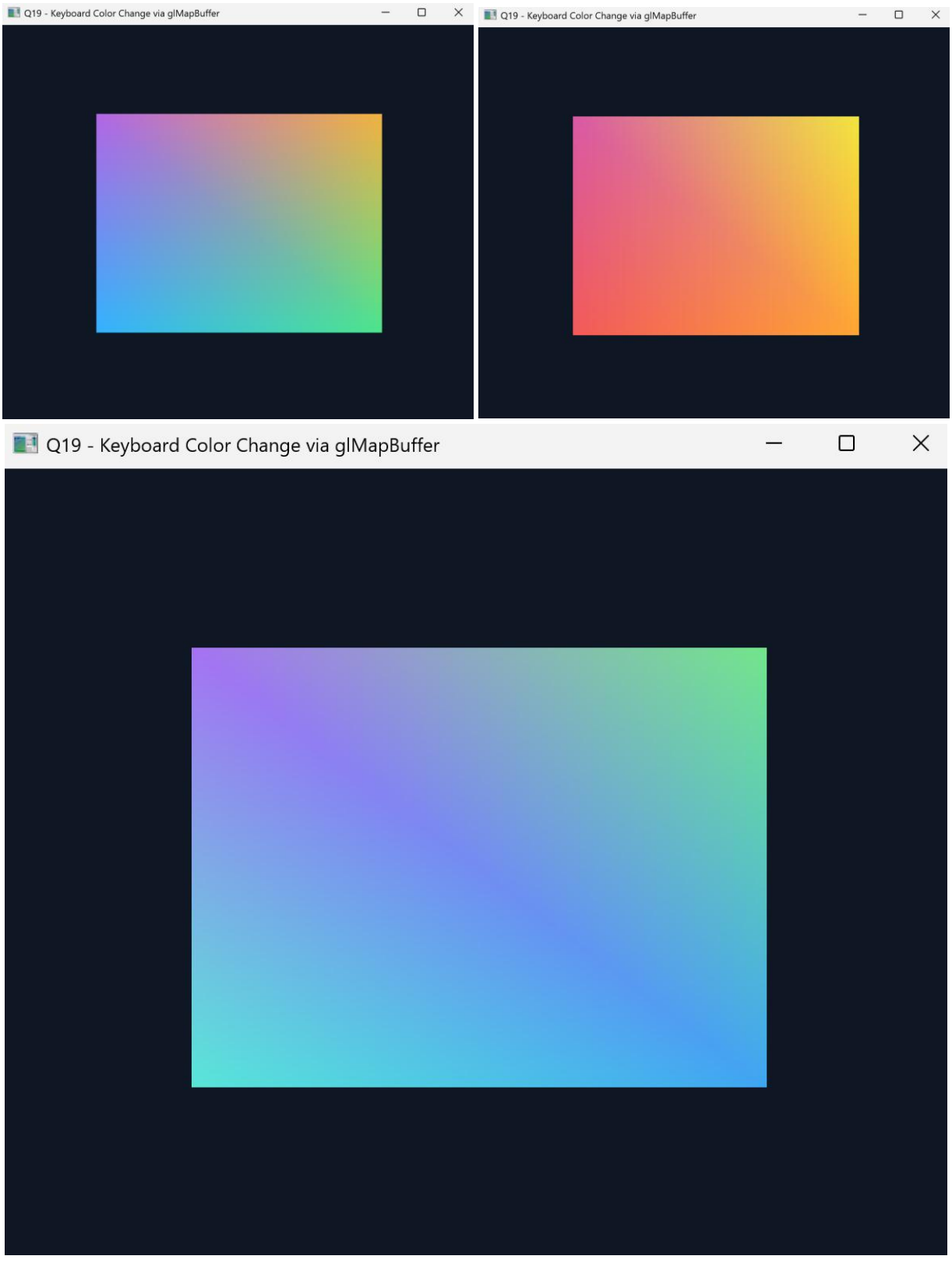
Exercise Q17 — Procedural 30-Dash Ring in One VBO [Hard]



Exercise Q18 — Interleaved + Indexed Hexagon [Hard]



Exercise Q19 — Keyboard-Triggered Color Change via glMapBuffer [Hard]



Exercise Q20 — Capstone: Animated Ship Over Indexed Ground [Hard]

